Sahaj Bhadja

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Summary

Software engineer with 2+ years building responsive applications, REST APIs, real-time mobile workflows, and testable production systems. Experienced with Java, React Native, Android/iOS delivery, Firebase persistence, debugging, and cross-functional development.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Architected multi-tenant application services on Azure SQL and Blob Storage for 500+ users, implementing row-level security, indexed persistence, and responsive access patterns that reduced dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant asynchronous workflows for 1,000+ monthly jobs with explicit state transitions, retries, failure isolation, and production monitoring, improving throughput by 3x while preventing cascading user-facing failures.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered React workflows and Flask REST APIs handling 10,000+ daily requests, integrating secure file ingestion, AWS RDS persistence, and responsive state updates while maintaining stable performance during peak usage.
- Implemented concurrent processing for 1,000+ uploaded files with validation, error isolation, structured logging, code reviews, and automated tests, reducing feedback turnaround by 60% for 150+ instructors.

USC AD Labs **Los Angeles, CA**
Machine Learning Researcher *January 2025 – May 2025*
- Built Python pipelines across 100,000+ trajectory records, producing reproducible metrics, logs, and visual summaries that improved debugging, benchmarking, and experiment traceability for interactive algorithmic system analysis.
- Documented experimental behavior, edge cases, and validation outputs for researchers, translating complex system signals into actionable findings while maintaining testable procedures and reproducible technical artifacts for team review.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Engineered cross-platform services with object-oriented design, Jenkins pipelines, automated builds, and 20+ PyTest and Mockito suites across Windows, Linux, and macOS, strengthening release quality and application maintainability.
- Optimized API and data-persistence workflows using AWS Glue, Amazon S3, SQL, indexing, caching, and pagination, improving application response times by 30% and production pipeline efficiency by 68%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs and React Native Android/iOS clients for booking, scheduling, payments, and status updates across 50,000+ records, integrating responsive interfaces with Firebase-backed persistence and real-time synchronization.
- Implemented Firebase transactions, Cloud Messaging notifications, session validation, and webhook-based payment updates, reducing manual handling by 70% while preventing double bookings and improving confirmation speed by 40%.

Skills

Languages: Java, JavaScript, TypeScript, Python, SQL, C++, Bash
Mobile & UI: React Native, Android, iOS, React, responsive UI, layouts, themes, styles, state management
Backend & Persistence: REST APIs, Flask, FastAPI, Node.js, Firebase, PostgreSQL, MySQL, Azure SQL, AWS RDS
Testing & Delivery: unit testing, integration testing, PyTest, Mockito, debugging, Git, code reviews, CI/CD, Jenkins
Engineering: object-oriented design, design patterns, networking, JSON, WebSockets, Agile, cross-functional collaboration, production support

Projects

HMIS | *Java, React, Appwrite, REST APIs, TailwindCSS*
- Built responsive appointment and patient-history workflows using Java, REST APIs, and authenticated persistence, enabling clinicians to access records 2x faster and coordinate 100+ appointments through user-centered interfaces.

FinNexus | *TypeScript, Next.js, MongoDB, WebSockets, Plaid*
- Developed responsive transaction interfaces supporting 50,000+ users and 1M+ records, integrating secure APIs, real-time state synchronization, persistent filtering, and token-based access controls across scalable application workflows.

CPU Scheduling Simulator | *C++, React, JavaScript, EJ-2*
- Implemented six scheduling algorithms with interactive controls, dynamic state updates, edge-case testing, and Gantt visualization across 1,000+ process combinations, reducing debugging time by 20% through deterministic simulations.